

Inference for Proportions: One & Two Samples

IMS Chapters 16 & 17

Why these two chapters go together

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- ▶ **Ch. 17 (two proportions):** the same machinery, extended to comparing two groups — which means one new wrinkle in how we compute the standard error.

Recap: one proportion, three ways

For a single proportion $\hat{p}$, you already know three ways to get a p-value or CI:

Method	Tool
Simulation (null true)	randomization / null simulation
Simulation (no null assumed)	bootstrapping
Mathematical model	$Z = \frac{\hat{p} - p_0}{SE}, \quad SE = \sqrt{p(1-p)/n}$

All three answer the same question — they just make different tradeoffs between speed, transparency, and required conditions (independence; for the normal model, also $np \geq 10$ and $n(1-p) \geq 10$).

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i Sample size for a desired margin of error

$$n = \left(\frac{z^*}{ME} \right)^2 p(1 - p)$$

If you have no prior estimate of p , use $p = 0.5$ — this is the most conservative choice, since $p(1 - p)$ is maximized at $p = 0.5$, giving the largest (safest) n .

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Two proportions

Comparing two groups: what changes?

Now we have two independent samples, and we care about the **difference** $p_1 - p_2$. The point estimate is simple: $\hat{p}_1 - \hat{p}_2$. The standard error is where it gets interesting — and it depends on *what question you're asking*.

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- ▶ **Building a confidence interval?** Use each group's own $\hat{p}$ — the **unpooled** SE.
- ▶ **Testing $H_0 : p_1 = p_2$?** Under that null, both groups are assumed to share one common proportion — so we should estimate that *shared* rate first, using **all** the data. This is the **pooled** SE.

The two standard errors

i Unpooled SE (for a confidence interval)

$$SE = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

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i Pooled SE (for a hypothesis test of $H_0 : p_1 = p_2$)

$$\hat{p}_{pool} = \frac{x_1 + x_2}{n_1 + n_2}, \quad SE = \sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

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Why the difference matters: under H_0 , we're assuming one shared truth — so we should estimate the SE consistently with that assumption, not by trusting each sample's own (possibly different) $\hat{p}$.

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$$n_1 \hat{p}_{pool} \geq 10, \quad n_1 (1 - \hat{p}_{pool}) \geq 10, \quad n_2 \hat{p}_{pool} \geq 10, \quad n_2 (1 - \hat{p}_{pool}) \geq 10$$

Case study: disaggregating tobacco use

A 2021 study (Rao et al.) examined current smoking rates among different Asian American subgroups, rather than lumping them together — a good example of why disaggregating categories can matter.

	Asian-Indian	Chinese
n	4,373	4,736
smokers	223	279
$\hat{p}$	0.0510	0.0589

Is the smoking rate among Asian-Indian Americans different from the rate among Chinese Americans?

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Is the smoking rate among Asian-Indian Americans different from the rate among Chinese Americans?

$H_0 : p_{Indian} = p_{Chinese}$ vs. $H_A : p_{Indian} \neq p_{Chinese}$ (two-sided, by default).

Running the test

$$\hat{p}_{pool} = \frac{223 + 279}{4373 + 4736} = \frac{502}{9109} \approx 0.0551$$

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$$SE_{pool} = \sqrt{0.0551(0.9449) \left(\frac{1}{4373} + \frac{1}{4736} \right)} \approx 0.00479$$

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Since $0.098 > 0.05$, we **fail to reject** H_0 — this sample doesn't provide convincing evidence of a difference between these two groups' smoking rates.

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Consistency check: the interval contains 0 — matching our “fail to reject” conclusion from the test. This will always line up, since both are built from essentially the same logic.

In code

```
# Two-proportion test/CI in R -- same function you've already used  
prop.test(x = c(223, 279), n = c(4373, 4736))
```

`prop.test()` handles the pooled/unpooled distinction internally — it's using the pooled SE to get the p-value, and (a version close to) the unpooled SE for the reported confidence interval, all in one call.

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- ▶ The one genuinely new wrinkle: **pooled vs. unpooled SE** when comparing two proportions — pool when your null hypothesis assumes one shared truth, don't pool when you're just trying to estimate two separate quantities.
- ▶ Next: this exact same one-vs-two-group structure repeats again for **means** — that's what the t-test extensions to two samples will look like.